

Evaluating the Impact of Integration of ESG Factors in Credit Ratings Using External Credit Assessment Institutions for Corporates

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Agenda

Research

Introduction

- a. About the Company
- b. Problem Identification
- c. Background
- d. Research Questions
- e. Methodology

Methodology Assessment

- a. Selection of CRAs
- b. Reviewing CRAs

Integration in IRB Modeling

- a. Defined Framework
- b. Scenario Analysis

Conclusion

- a. Contribution to Research
- b. Limitations & Future Work

Results & Discussion



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About the Company



Largest Dutch bank¹ and leading European bank offering a broad range of services to its clients in retail and wholesale banking

Amsterdam, the Netherlands

Size: ~57.000 employees

Team: Integrated Risk, Wholesale Banking, Model Development

Involved in thesis with:

- a. Thesis
- b. Supervision

¹ ING Group total assets for Sept. 2024 were \$1153.997 billion.



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Problem Identification

The growing demand for sustainable investment and the integration of ESG principles in risk management highlight the need for financial institutions to better understand how **ESG factors influence credit ratings**

Regulatory bodies emphasize the need for banks to consider ESG risks in their **credit risk models**

However, significant variations in how **External Credit Assessment Institutions (ECAIs)** embed ESG factors create challenges for aligning **internal risk models** with external rating methodologies



Background

The growing demand for sustainable investment and the integration of ESG principles in risk management highlight the need for financial institutions to better understand how **ESG factors influence credit ratings**



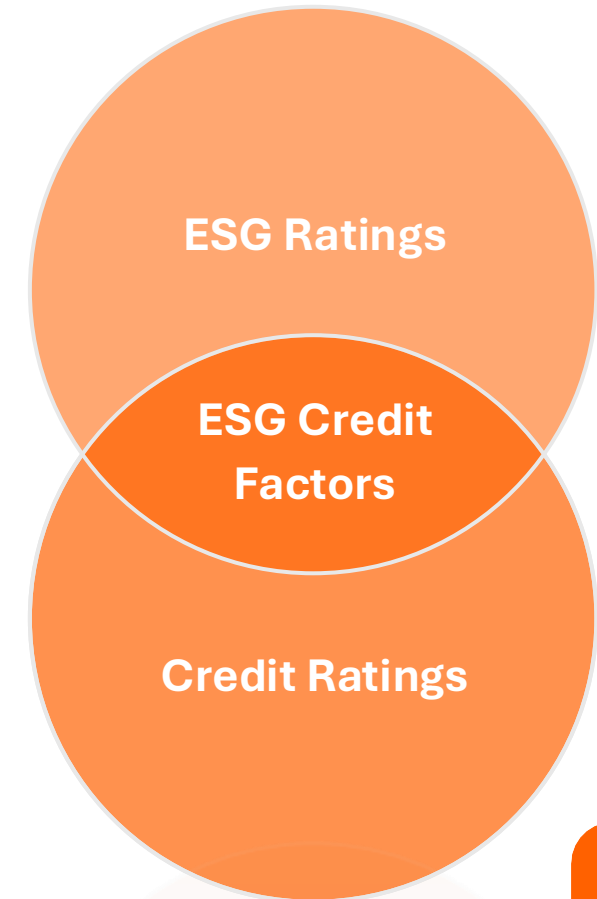
Background

ESG and credit ratings are both data-driven assessments by CRAs:

- a. **ESG ratings** reflect entity's sustainability profile, i.e., its impact on society and the environment and exposure to sustainability related risks
- b. **Credit ratings** evaluate an entity's creditworthiness and ability to meet financial obligations

Purpose: ESG ratings focus on **sustainability and governance**, while credit ratings assess **financial risk**

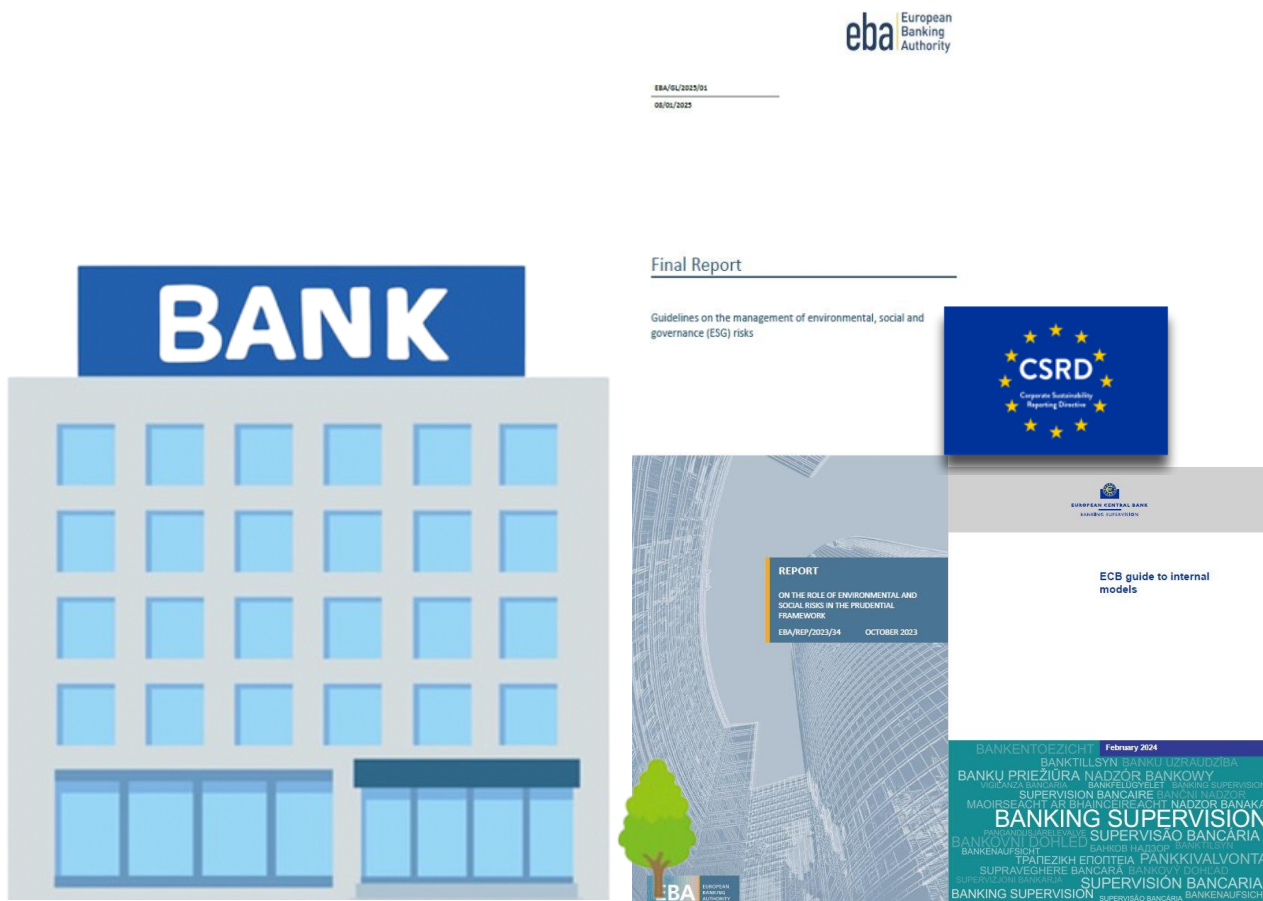
Strong creditworthiness does not necessarily correlate with strong ESG credentials and vice versa



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Background

Regulatory bodies emphasize the need for banks to consider ESG risks in their **credit risk models**



Various regulations exist for **credit risk** in the realm of ESG



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Background

Credit risk refers to the possibility that a borrower or counterparty will **fail to meet their obligations** to repay a loan¹ in accordance with the agreed terms

Fundamental concern for financial institutions, lenders, and investors, as it directly impacts the likelihood of recovering the full value of a loan or investment

Risk-based capital requirements are **minimum capital requirements for banks set by regulators**. The requirements act as a cushion to **protect a company from insolvency**.



¹ For most banks, loans are the largest and most obvious source of credit risk



Background

Banks may use their **own internal models** to estimate key risk parameters under the IRB approach

However, before these models can be used to calculate risk-weighted assets, they must first receive initial **approval from ECB Banking Supervision**

For measuring credit risk, banks can use various approaches:

- Standardized Approach (SA)
- Foundation Internal-Rating Based Approach (FIRB)
- Advanced Internal-Rating Based Approach (AIRB)

STANDARDIZED APPROACH (SA)

$$\text{Credit Risk Capital Requirement} = \text{Constant} \times \text{Exposure} \times \text{Risk Weight}$$

FOUNDATION INTERNAL-RATING BASED APPROACH (FIRB)

$$\text{Credit Risk Capital Requirement} = \text{Constant} \times \text{EAD} \times \text{PD} \times \text{LGD} \times \text{M}$$

ADVANCED INTERNAL-RATING BASED APPROACH (AIRB)

$$\text{Credit Risk Capital Requirement} = \text{Constant} \times \text{EAD} \times \text{PD} \times \text{LGD} \times \text{M}$$

Parameters estimated by financial institutions

EAD = exposure at default, LGD = loss given default, M = effective maturity, Constant = Risk-based multiplier defined



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Background

Risk weights are a fundamental aspect of the SA, **representing the credit quality of borrowers** or counterparties

The process uses **credit quality steps (CQS)**, a standardized scale that translates credit ratings (e.g., AAA, AA, A) into categories ranging from CQS 1 (highest credit quality) to CQS 6 (lowest credit quality), which results in a respective risk weight

Regulators assign risk weights based on external credit ratings provided by **External Credit Assessment Institutions (ECAIs)**, or Credit Rating Agencies (CRAs)



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Background

Role in crisis. Credit rating agencies played a prominent role in the 2007-2008 financial crisis by underestimation the risk

Regulatory steps. Since then, several steps have been taken in the EU to enhance the regulatory and supervisory frameworks for credit rating agencies (CRAs)

Established frameworks:

- a. Credit Rating Agencies Regulation
- b. Exhaustive list of registered and authorized CRAs in the EU at any point in time (**ECAIs***)
- c. Mapping of CRA assessments to the credit quality steps provided in the CRR



* **External credit assessment institutions (ECAIs)** are defined in CRR, Art. 4(98) as a credit rating agency (CRA) registered or certified according to the CRA Regulation, or a central bank issuing credit ratings which are exempt from the CRA Regulation



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Background

Role of ECAIs in SA: the application of risk weights under SA is based on classification of exposures to exposure class and credit quality; the credit quality of exposure is usually determined by reference to the ECAI

Role of ECAIs in IRB: under certain conditions, PD models can be calibrated by mapping internal grades to those used by an ECAI or a similar organization

No mechanical reliance on ECAI is permitted: institutions must perform own due diligence

Credit Quality Step	Fitch's assessments	Moody's assessments	S&P assessments	Corporate	Institution (includes banks)			Sovereign
					Sovereign method	Credit Assessment method		
						Maturity > 3 months	Maturity 3 months or less	
1	AAA to AA-	Aaa to Aa3	AAA to AA-	20%	20%	20%	20%	0%
2	A+ to A-	A1 to A3	A+ to A-	50%	50%	50%	20%	20%
3	BBB+ to BBB-	Baa1 to Baa3	BBB+ to BBB-	100%	100%	50%	20%	50%
4	BB+ to BB-	Ba1 to Ba3	BB+ to BB-	100%	100%	100%	50%	100%
5	B+ to B-	B1 to B3	B+ to B-	150%	100%	100%	50%	100%
6	CCC+ and below	Caa1 and below	CCC+ and below	150%	150%	150%	150%	150%

ECAIs suitable for IRB

ECAIs suitable for SA

Requirements for ECAIs under SA are to be considered for IRB

Not all ECAIs used under SA are suitable for IRB modelling due to stricter requirements for IRB¹

¹ The Basel Committee on Banking Supervision (BCBS) specifies that banks must demonstrate the accuracy and consistency of their internal rating systems, which involves rigorous validation processes



Background

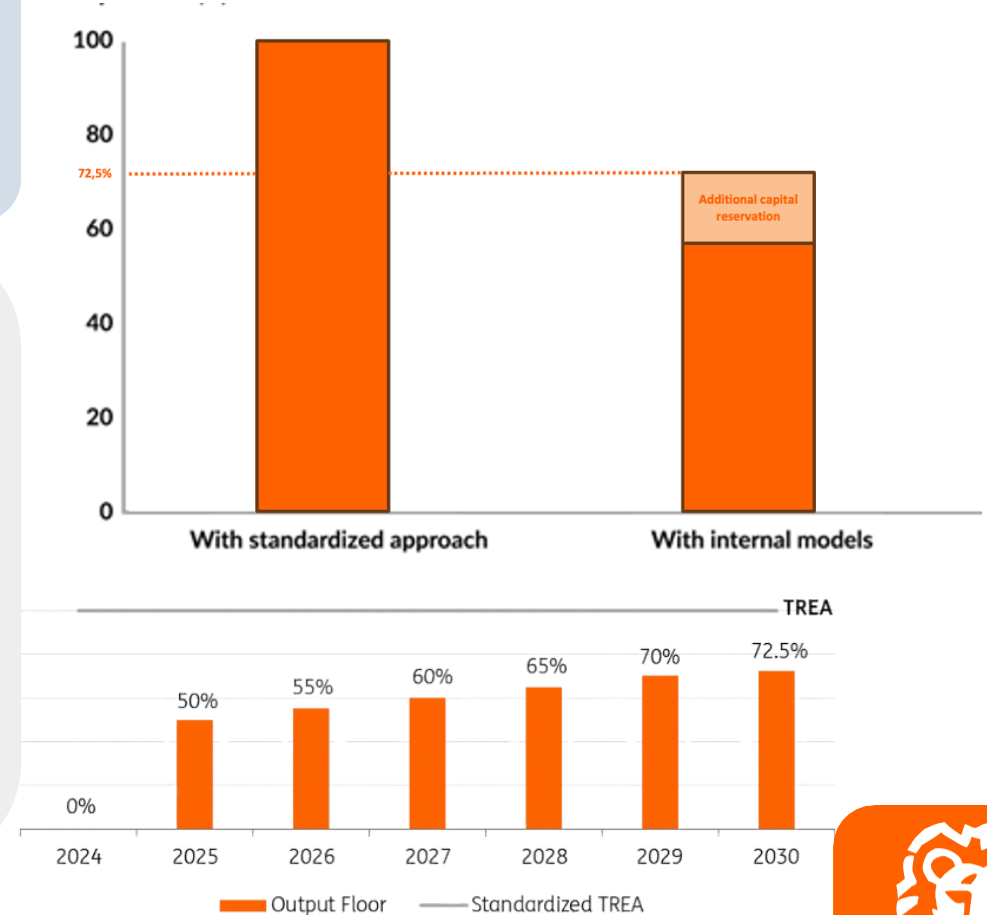
Output floor: According to CRR, Art. 92, banks must compute an output floor for capital requirements, defined as a percentage of the capital requirements under SA. Will eventually be **72.5%** of SA



Example

If *Company ABC* has a risk weight under SA assigned of 100%, this would mean the risk weight for internal models should be at least 72.5%.

One can see that the 72.5% risk weight requirement is higher than the calculated risk weight using internal models, meaning **additional capital reservation** is required



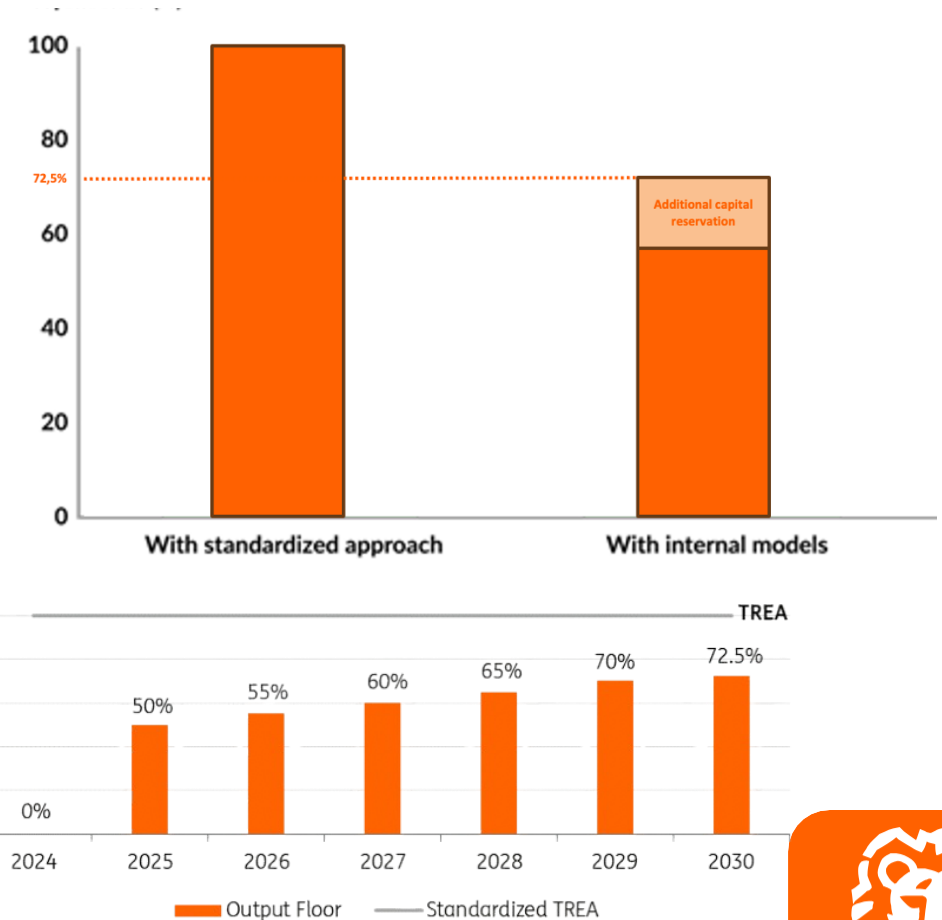
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Background

Unrated exposures issue: The risk weight under SA for corporate exposures without a credit assessment by a nominated ECAI, is rather high¹. Consequently, applying the output floor may result in **capital allocation that is misaligned** with the underlying risk

A **transitional period** is specified until December 2029. This period should be used to **significantly expand the availability of external credit ratings** for corporates.

After this period, institutions should be able to refer to assessments by nominated ECAIs to calculate own funds requirements for a significant part of their corporate exposures



¹ According to CRR, Art. 122(2) a risk weight of 100% must be assigned for unrated corporate exposures. However, pursuant to CRR, Art. 465(3), until December 2032 a risk weight of 65% can be assigned instead if IRB PD for the corporate obligor is not greater than 0.5%



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Background

However, **significant variations in how External Credit Assessment Institutions (ECAIs) embed ESG factors** create challenges for aligning internal risk models with external rating methodologies



How do I map my internal credit grades to external ratings embedding ESG when ECAI methodologies vary?



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Background

Potential ways to include ESG into IRB models:

Calibration: incorporation into the calibration levels can be provided through the mapping to external credit ratings (provided by ECAs) that accounts for ESG risks

Implementing ESG in risk drivers: Obligors are generally subject to multiple ESG disclosure obligations¹ or follow recommendations on such disclosers². Information stemming from ESG disclosures forms a basis for the risk drivers that can be tested in the models

1 – SFDR, CSRD, etc.

2 – TCFD, TNFD, ISSB, etc.



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Frameworks

Looking into individual frameworks described by ECAs



Time Horizon

Assessing how ECAs are integrating different time horizons for ESG risk



CSRD Mapping

Looking into integration of ESG topics mentioned in CSRD



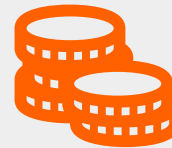
Industry

Looking whether ECAs consider different industry characteristics in assessment



Materiality

Shows what ECAs are prioritising in terms of ESG risks



Costs

Looking which costs are considered in ECA methodologies



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Selection of CRAs

CRAs have established methodologies to provide credit ratings for various counterparties in various geographies

Currently there are **24*** registered and certified CRAs

* All branches of Moody's are counted as a single CRA

Selected CRA methodologies with a dedicated framework for assessment:

1. S&P Global Ratings Europe Limited[#]
2. Moody's Investors Service[#]
3. Fitch Ratings Ireland Limited[#]
4. DBRS Rating GmbH
5. Scope Ratings GmbH
6. Ethifinance Ratings
7. HR Ratings de México, S.A. de C.V.



Together accounting for **over 90%** market share

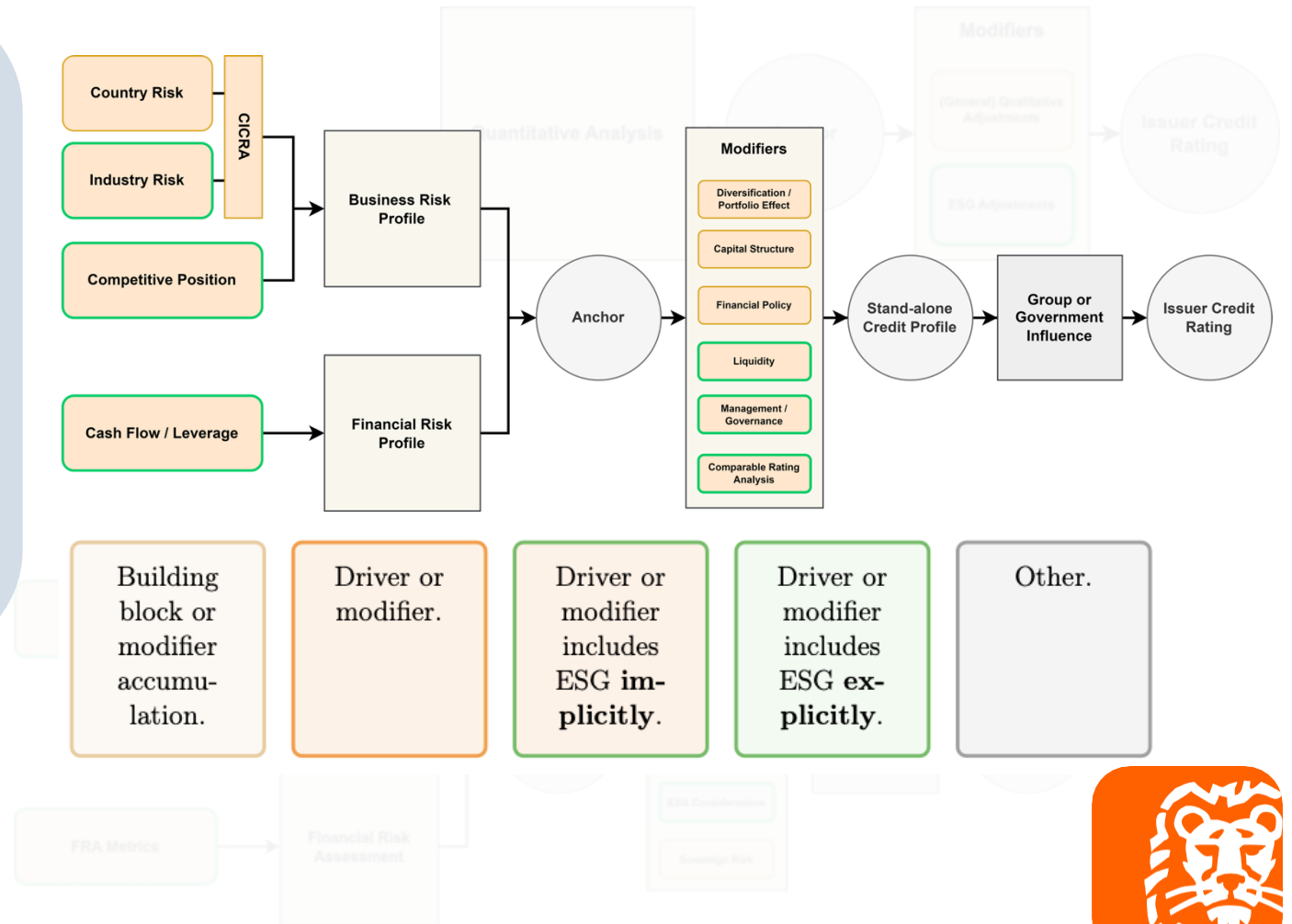
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Reviewing CRAs: Frameworks

Structured frameworks: ECAs use structured frameworks with recurring components like:

1. Business & financial risk profiles
2. Anchor ratings
3. Modifiers

CRAs integrate ESG factors quantitatively and qualitatively via risk drivers, building blocks, or modifiers. The **ESG integration approaches vary** across CRAs



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Reviewing CRAs: Costs



- a. **Carbon pricing costs:** Taxes, offset expenses, compliance costs related to GHGs
- b. **Physical risk costs:** Damage from extreme weather, remediation expenses, and adaptation measures
- c. **Transition costs:** Investments in low-carbon technologies and green transformation initiatives
- d. **Stranded asset risks:** Loss of asset value due to obsolescence in carbon-intensive industries

E



- a. **Labor costs:** Increased wages, benefits, and retention strategies
- b. **Health and safety costs:** Compliance with safety regulations and training
- c. **Productivity losses:** Impact of strikes or hostile work environments
- d. **Litigation costs:** Legal disputes arising from social or labor issues

S



- a. **Regulatory compliance:** Expenses related to governance reforms, non-compliance, anti-corruption measures, and public disclosures
- b. **Fines and penalties:** Costs from governance failures or legal non-compliance
- c. **Operational inefficiencies:** Challenges arising from conflicts of interest or structural complexities

G



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Reviewing CRAs: Time Horizon

S&P Global
Ratings

Visibility of ESG factor

1

Clear within financial
forecast horizon

2

Clear beyond financial
forecast horizon

3

Unclear

Incorporation to credit rating*

Incorporated quantitatively:
factored into financial forecast
(FRP)

Incorporated qualitatively:
forward-looking assessment of
industry risk and competitive
position of the entity (BRP)

Not incorporated: considered for
rating monitoring only

Examples

Costs of greenhouse gas emission,
waste management, other pollution,
health and safety

ESG **policy milestones** such as approved
regulatory decisions

Unexpected drastic shifts in technology,
customer behavior,
extreme climate events,
potential public policy changes

* The incorporation takes place only where ESG factor materially affects creditworthiness



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Reviewing CRAs: Materiality

EthiFinance

For the assessment of entity's competitive position, EthiFinance has developed a scorecard that is based on **18 scorable ratios / indicators**. The ESG indicators have been selected with **financial materiality** in mind and are **applicable to entities in all industries***.

FP can be either adjusted positively or negatively (-1 notch or +1 notch)

They are categorized under E, S, and G and have been selected to evaluate the entities around these themes.

E

1. **Existence of an environmental management system (EMS)** and share of activities covered by an external certification (e.g., ISO 14001, EMAS)
2. **Total Energy consumption** (MWh/m€)
3. **Greenhouse gas emissions, Scope 1 (direct emissions) and Scope 2 (indirect emissions from purchased energy)**(tCO₂e/m€)
4. **Water consumption** (m³/m€)
5. **Waste rate** (t/m€)

G

15. **Number of independent board members**
16. **Public disclosure of a formalized Business Code of Conduct and Corruption Policy**
17. **Assessment and prioritization of the group's ESG issues**
18. **Separation of the roles of CEO and Chair of the Board**

S

6. **Average number of training hours per employee** (total training hours / total workforce)
7. **Permanent employee turnover rate** (number of permanent employee departures (FTE) / total workforce (FTE))
8. **Existence of an HSS management system** (health, safety, security) and share of activities covered by an external certification (ISO 45001)
9. **Absenteeism rate** for illness and work accidents
10. **Accident frequency rate** (number of accidents with lost days × 1,000,000 / number of hours worked)
11. **Share of women in management positions**
12. **Responsible purchasing policy including social criteria**
13. **Responsible purchasing policy including environmental criteria**
14. **Existence of a quality management system (QMS)** and share of activities covered by an external certification (e.g., ISO 9001)



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* Except waste and water consumption that are not applicable to some industries

Reviewing CRAs: Industry

EthiFinance

For the assessment of industry risk, EthiFinance has developed a **sector heat map** that assigns scores to different environmental- and stakeholder-related topics

Scores are evaluated on a scale from 1 to 5, which would affect the industry risk score:

- a. **1** = Lowest ESG-related impact
- b. **5** = Highest ESG-related impact

Scores are determined for each sector based on **double materiality** assessment. Some sectors are affected controversially while other benefit from ESG trends

Sector	Environmental factors				Stakeholders			Global risk
	Climate	Resources	Pollution	Biodiversity	Suppliers	Consumers	State, regions, communities	
Consumer goods (both branded and private): processed food, household & personal products, consumer durables & apparel	2,4	2,4	3,9	3,0	3,0	3,4	3,1	3,4
Oil, gas, coal, energy equipments, electricity & gas utilities	4,8	4,0	4,8	4,5	2,8	3,5	4,4	4,4
Renewables, water utilities, multi utilities	1,0	2,1	1,0	1,0	1,8	1,0	1,0	1,7
Agribusiness	3,5	3,8	3,8	4,0	3,5	3,8	3,4	3,8
Beverages	1,0	2,0	3,0	2,0	2,0	4,1	4,4	3,5
Healthcare equipment & services	1,0	1,0	2,5	1,5	2,8	2,4	3,8	2,9
Hotels and leisure	2,9	2,5	2,8	3,0	2,0	3,2	2,8	2,9
Capital goods: aerospace, defence, conglomerates, building products and machinery	3,5	4,0	3,8	3,0	3,8	3,0	1,0	3,6
Auto constructors	4,0	4,0	4,8	4,0	3,8	3,8	3,1	4,3
Auto component manufacturers	3,5	3,8	3,0	3,0	3,8	3,4	3,1	3,6
Environmental services	1,0	2,0	1,0	1,0	2,0	1,8	2,0	1,8
Information technology: hardware equipment, electronic instruments, semi-conductors and semi-conductors equipment	3,0	3,4	2,8	3,0	3,4	3,4	1,8	3,2
Infrastructures and construction & engineering	3,0	3,8	3,5	3,0	2,0	2,8	2,5	3,3
Materials and chemicals	3,8	4,8	4,8	4,0	3,5	2,8	2,0	4,2
Media and telecommunications	2,0	1,0	1,8	1,0	1,0	2,8	2,8	2,3
Real estate developers	3,0	3,8	2,8	3,0	2,8	1,8	2,9	3,3
Services and retailing: food & staples retailing, general retailing, commercial & professional services, software services	2,0	3,0	2,8	2,0	4,0	3,9	1,0	3,3
Transportation cyclical (airlines, road and marine transport)	4,8	4,8	4,8	2,0	4,0	4,0	3,0	4,3
Railways	1,8	2,8	2,0	1,0	3,0	3,0	1,8	2,6

ESG in Credit Ratings

EthiFinance

ESG controversies: comparison of entity's ESG public communications and actions.

Controversies score evaluates the **potential financial impact** of controversies. The score considers past news, lawsuits, and reputation damages with future implications.

Controversies score results in **rating downgrades**:

- **Score 5:** downgrade by **2 notches**
- **Score 4:** downgrade by **1 notch**

To avoid double counting, no rating downgrade is performed if ESG company score is between 4-5

Score	Controversies / Definition
1	News or events that do not constitute a real issue but indicate a weakness in company operations. Requires monitoring .
2	News/events that do not pose a material financial risk but temporarily impact reputation . Requires monitoring .
3	Unexpected news/event that may affect reputation, organization, or financials in a manageable way. Recurring events could escalate to a significant issue.
4	A series of unexpected events leading to a reassessment of the business model . Potential for significant impact on growth or debt metrics .
5	A series of major events leading to a significant reassessment of the business. Expected to permanently affect growth or debt metrics.



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Reviewing CRAs: Commonalities & Differences

Commonalities Across CRAs

- a. Inclusion of ESG factors:** All CRAs address climate change, social responsibility, and governance as integral parts of credit assessments
- b. Integration of qualitative judgments:** CRAs supplement quantitative data with expert judgment to evaluate ESG risks that lack standardization or clear visibility
- c. Forward-looking analysis:** Most CRAs incorporate dynamic forecasts to evaluate the evolving impact of ESG risks over time
- d. Sector-specific approaches:** Tailored methodologies reflect industry-specific ESG risk
- e. Challenges with data and standardization:** Limited access to high-quality, consistent ESG data is a barrier acknowledged by all CRAs

Differences Among CRAs

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Reviewing CRAs: Commonalities & Differences

Commonalities Across CRAs

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Differences Among CRAs

- a. **Disclosure approaches:** S&P and Moody's explicitly define ESG drivers in their methodologies, while Fitch uses relevance scores to highlight indirect impacts
- b. **Materiality frameworks:** Scope and EthiFinance apply double materiality, considering both financial and sustainability impacts, unlike single materiality approaches by S&P or Moody's
- c. **Time horizon focus:** S&P prioritizes short- and medium-term risks, whereas DBRS includes long-term ESG impacts over a rating lifecycle of up to 30 years
- d. **Cost attribution:** Moody's highlights labor and productivity costs, while EthiFinance focuses on operational costs like waste management
- e. **Transparency and methodology:** Scope and EthiFinance provide dynamic materiality matrices, whereas DBRS emphasizes clarity through visual flowcharts in their methodologies



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Integration into IRB Modeling: Defined Framework

Rating adjustments done via a structured approach set up:

- a. Asking questions to see how ESG factor is influencing the borrower
- b. Materialities and time horizons are assessed for the respective ESG factors to see whether it falls within regulatory time horizons
- c. Finally, based on a classification into either business or financial risk profile, a choice is made for an integration as a **driver/ override of driver** or **rating override**

ESG Risk

Questions:

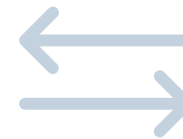
- a. Would ESG risk affect risk profile?
- b. Would it affect brand strength or reputation?
- c. Does factor affect earnings or cashflow?

Materiality

Time Horizon



Driver



Modifier



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Integration into IRB Modeling: Scenario Analysis

A **scenario analysis** was conducted to provide modelers with more practical examples

- a. Defined through CSRD
- b. All features from the framework are used within the analysis

By following these guidelines, risk modelers will be able to **systematically address** different ESG-related challenges

Decisions can be taken in line with evolving regulatory frameworks

Climate Change (E1)

Recommendation 1 A utility company reliant on coal energy is exposed to the material risk of stranded assets as policies and market trends shift toward renewable energy sources. The IRB model could incorporate this material risk by adjusting the asset depreciation rates and projecting future cash flow reductions. This could result in a higher Probability of Default (PD), especially when assessing the company's short to medium-term financial outlook, considering the immediate costs of regulatory shifts and market demand changes.

Recommendation 2

The same utility company shows limited readiness to transition to low-carbon operations, highlighting a longer-term ESG risk. This lack of transition readiness should be captured as a qualitative modifier in the IRB model. The assessment would include factors such as the company's plans to adapt, the potential costs of delayed transition, and the risk of future regulatory and market disruptions. By including this modifier, the model reflects the increasing long-term risk in the company's credit rating, particularly as ESG expectations evolve over time.

Recommendation 3

For high-carbon industries, such as this utility company, an ESG overlay could be applied to account for broader sectoral risks. These could include long-term regulatory pressures, such as carbon pricing or emission restrictions, and shifts in market demand due to decarbonization trends. The overlay would factor in both the short-term costs of regulatory compliance and the long-term costs associated with transitioning to sustainable practices. This approach helps model the cumulative impact of ESG risks across industries, ensuring that sectoral risks are properly integrated into the IRB model.



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Conclusion

“How do ESMA-registered External Credit Assessment Institutions (ECAIs) integrate ESG factors into their credit rating methodologies?”

ESG factors are **inconsistently defined** across ECAIs

Some ECAIs **treat ESG as standalone criteria** (modifier), while others try to **integrate them into building blocks** like operational (business risk profiles) or financial risks (financial risk profiles)

Including ESG factors **impacts the comparability and consistency** of assessments among ECAIs

Additionally, CRAs provide **insufficient disclosure** about how ESG factors influence their ratings, reducing stakeholders' ability to evaluate and compare methodologies effectively

Lack of standardization in how ESG factors are defined, weighted, and incorporated

A lack of uniformity leads to inconsistent frameworks, but gives **room to own interpretation**

Conceptual framework is proposed to help IRB modelers consider ESG risks

The thesis proposes a conceptual framework that focuses on features identified through the methodologies of CRAs



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Contribution to Research

Provides a **landscape overview** for ESG integration of various ECAs

Identifies key inconsistencies in ESG integration across ECAs, **evaluates the impact** on credit rating comparability and transparency, and **proposes a set of heuristics** to guide financial institutions

Literature only looks at ESG from a very high-level perspective. This thesis provides a **more practical approach** to the integration possibilities of ESG in risk management



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Limitations & Future Work

Limitations

- a. **Short timeframe:** The research was conducted over a relatively brief period, limiting the depth of data collection and analysis
- b. **Regulatory landscape evolution:** Given the dynamic nature of ESG-related regulations and credit rating methodologies, it is likely that certain aspects will become outdated within a year as new regulations and practices emerge
- c. **Reliance on available disclosures:** The study heavily relied on publicly available ESG and credit rating disclosures from ECAs
- d. **Evolving ESG definitions and metrics:** The definitions for ESG are still evolving, influencing consistency and comparability of the ESG-related assessments across different ECAs
- e. **Subjectivity in ESG integration evaluation:** ESG factor integration often involves subjective judgments by rating analysts and the author for reading the documentation

Future Work

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Limitations & Future Work

Limitations

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Future Work

- a. Comprehensive analysis of all ECAs defined by ESMA:** Expand the scope of research to include all ESMA-registered ECAs, providing a complete overview of ESG integration practices across the regulatory landscape
- b. Incorporation of additional asset classes:** Investigate how ESG factors are integrated into credit ratings for asset classes beyond corporates, such as specialized lending, financial institutions, and sovereigns
- c. Industry-specific ESG assessments:** Explore how ESG factor integration varies across industries, identifying sector-specific challenges, opportunities, and materialities
- d. Exploration of ESG factor materiality by asset class:** Analyse the relative importance of ESG sub-factors, such as transition risk or governance practices, across different asset classes



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ESG Incorporation in Credit Ratings



The sustainability topics addressed by CRAs in credit ratings are broadly consistent with those under **CSRD**, but the level of granularity varies:

- a. **Climate change (E1)**: covered by all CRAs
- b. **Pollution & Biodiversity and ecosystems (E2, E4)**: broadly covered, but Scope and HR considerations lag
- c. **Water & Marine resources (E3)**: only Moody's, Fitch, and EthiFinance address the topic
- d. **Own workforce & Workers in the value chain (S1, S2)**: own workforce is considered, but value chain labor risks are less covered
- e. **Business conduct (G1)**: business conduct is consistently integrated

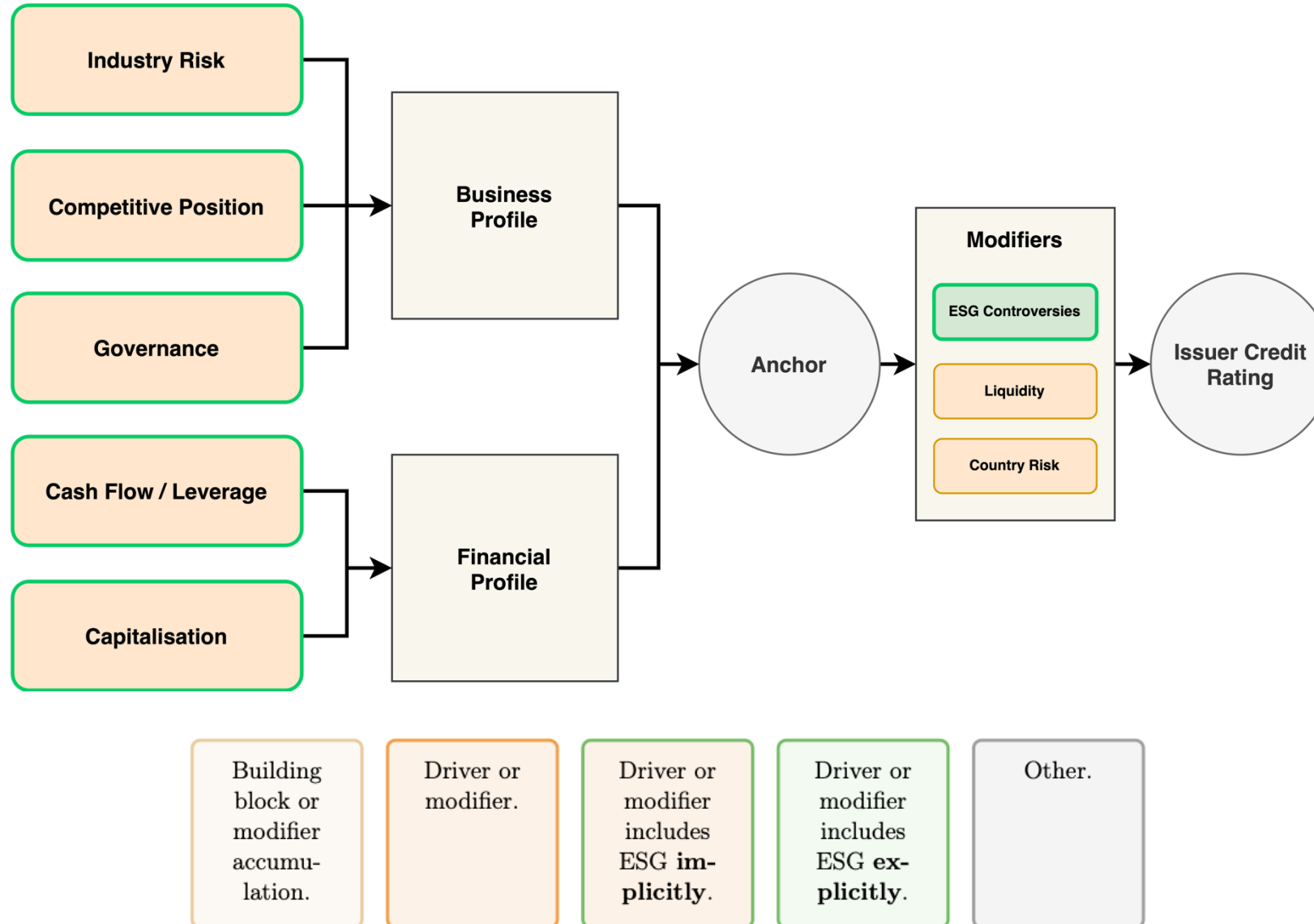
Category	Topic	S&P	Moody's	Fitch	DBRS	Scope	EthiFinance	HR
E	Climate change	✓	✓	✓	✓	✓	✓	✓
E	Pollution	✓	✓	✓	✓	✗	✓	✗
E	Water and marine resources	✗	✓	✓	✗	✗	✓	✗
E	Biodiversity and ecosystems	✓	✓	✓	✓	✓	✓	✗
E	Resource use and circular economy	✗	✓	✓	✓	✓	✓	✗
S	Own workforce	✓	✓	✓	✓	✓	✓	✓
S	Workers in the value chain	✗	✓	✓	✗	✗	✗	✗
S	Affected communities	✓	✓	✓	✓	✓	✓	✓
S	Consumers and end-users	✓	✓	✓	✓	✓	✗	✗
G	Business conduct	✓	✓	✓	✓	✓	✓	✓



do your thing

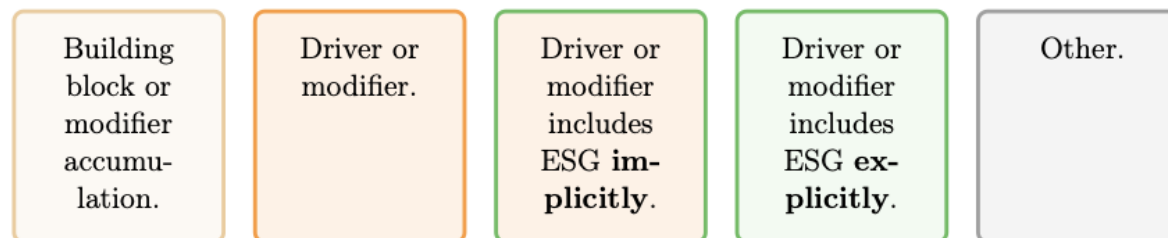
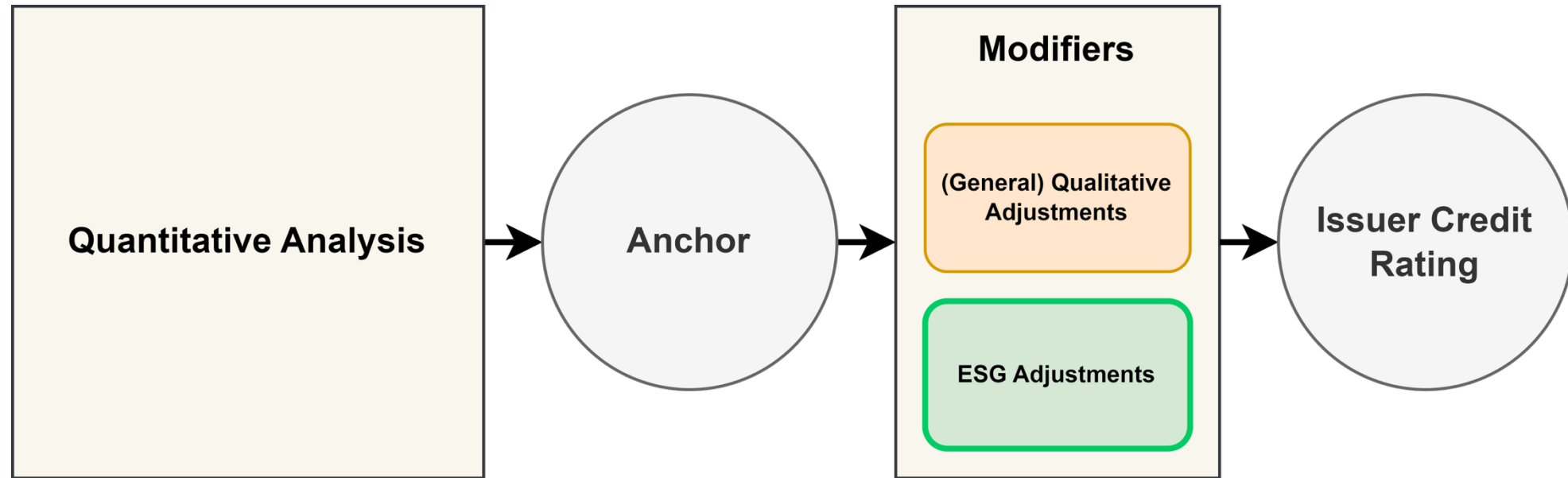
Framework Example: EthiFinance Ratings

EthiFinance



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Framework Example: HR Ratings



do your thing